

Assignment 1

Is the Brain Like a Computer?

The question of the brain's similarity to a computer must be addressed on multiple levels to arrive at a coherent answer, each more abstract than the last. First, the basic physical structure of both systems must be considered; second, the activities of each system; and third, the role that semantics play in each system (and how these roles may be related). The existence of the conscious mind as a property of the brain is the crux of this question, and, because of computational theory's failure to address it adequately, the brain from which it blooms is not like a computer in a sense that carries any philosophical import.

Computers and brains share some similarities in physical structure. Both systems operate using electricity, which transmits binary information – in brains, a synapse is either firing or it isn't, and in computers, a binary digit is either 0 or 1, with no variation. These structures abstract into networks that can perform complex tasks, and, according to a materialist view of the brain, these networks can not only represent but *be* the subjective human experience. Paul Churchland's reductive physicalism argues that the prior statement is the fact of the matter, and that every subjective mental state can, in theory, be explained by physical, observable brain states. Daniel Dennett, too, argues that mental states are properties of physical matter, and therefore that human consciousness is an illusion of the brain's biology. These views seem reasonable when one considers a philosophy like connectionism in tandem – its more sophisticated computational view of the brain, in which nodes make up networks of greater and greater complexity, makes it easier to grasp how a purely physical, binary system can become something with as much complexity and detail as subjective consciousness (whether human or machine).

Of course, if materialism ascribes the mind to the physical brain, then its activities must also be explained in purely physical terms. This is possible with a computer – it receives input, performs a computation according to a predetermined program, and produces output. Putnam's functionalism holds this process to be true for the brain as well, with the brain performing computations that are of no import except in their output – a black box. Such a simple explanation for the brain and its property of consciousness is not possible, however, if one considers Searle's rebukes of Dennett and Putnam. According to Searle, both philosophers fail to account for consciousness in their theories - Dennett denies the wealth of subjective, conscious experience data that humans produce, while Putnam's model doesn't account for consciousness as a property of the brain's biology. While the brain indeed takes input, performs computations, and produces output, its constant creation of a conscious mind cannot be adequately explained by a purely computational philosophy.

In his rebuke of Dennett, Searle raises another critical point – computers use syntax just like human beings, but they lack both semantics and intentionality. Husserl's phenomenology is influential here. He argued that the mind is not passive, but constantly seeking and actively experiencing the world and assigning meaning to those experiences. Fodor's mental "representations" - the main way humans access reality in his philosophy - are inextricable from semantics as well. Computers are passive – they have static intentionality determined by their programming, and no semantics except for those that humans ascribe to them. Exercises like the Turing test, then, don't carry any import. If a machine can "think", then it is the human ascribing

that meaning to what the machine is doing – the machine itself applies no such meaning to its activity.

Computers offer a useful way to view many machinations of the brain, but the analogy stops short of solving philosophical problems about the mind. So, while the brain and the computer share many similarities in structure and basic function, the deeper questions of meaning and consciousness, as well as the problem of the relationship between the mind and body, remain unsolved by a comparison of the two systems, especially when abstracted to consider the divergent functions and the role of semantics in each system. Therefore, the brain is not like a computer in any meaningful sense.